

S.No	Electrical Components	Mechanical Components	Objective of the robot	Video link
1	Arduino Nano, flex sensors, 5*10 kohm resistors	3D printed fingers, palm frame, nylon tendons	To detect finger bending and reproduce the same movement on a robot hand	https://www.youtube.com/watch?v=mOlshih33gg
2	Arduino, Nano, nRF24L01 modules, flex sensors	3D printed hand, MG996R servos, string tendons	To control a robotic hand wirelessly using human hand gestures	https://www.youtube.com/watch?v=FW0VScrXpAY
3	Arduino uno, servo motors, flex sensors	PLA hand, nylon strings, wearable glove	To demonstrate a low cost robotic hand that follows finger movements	https://www.youtube.com/watch?v=KqeZ_P5rO6E
4	Arduino Nano, MPU6050, flex sensor	3D printed robotic arm and gripper	To combine gesture and motion sensing for robotic movement	https://www.youtube.com/watch?v=Py3qWZeAV7E
5	Arduino Nano, flex sensors, servo controller	PLA fingers, hinge joints, elastic bands	To achieve controlled bending of multiple robotic fingers	https://www.youtube.com/watch?v=Fvg-v8FPcig
6	Arduino Nano, wireless transmitter/reciever, servos	3D printed palm, finger links, braided cable	To transmit finger position information wirelessly to a robotic hand	https://www.youtube.com/watch?v=QOyghUxLdqE
7	Arduino, flex sensors, rechargeable battery	Lightweight PLA hand, tendon wires, finger joints	To develop a portable hand that responds to finger bending	https://www.youtube.com/watch?v=f7hHvrXVjak
8	Arduino Nano, flex sensor, servo motor	3D printed gripper, support frame, elastic cords	To demonstrate sensor based control of a robotic gripper	https://www.youtube.com/watch?v=cbWCoEaQ_FY
9	Arduino, 5 servo motors, flex sensors	PLA hand, nylon tendon cables, glove mount	To reproduce five basic finger movements for human robot interaction	https://www.youtube.com/watch?v=RLTetVgm4bl
10	Arduino, nRF24L01, flex sensors, servo control	3D printed hand and forearm, strings and fasteners	To create a complete wireless gesture controlled robotic hand prototype	https://www.youtube.com/watch?v=yyI0KVxzZIM